

Answer: A

22



$$\lambda_1 = 4L$$



$$\lambda_2 = 4L/3$$

Resonance occurs at wavelengths $= 4L, \frac{4L}{3}, \frac{4L}{5}, \frac{4L}{7}$

Resonance occurs at frequencies $= \frac{v}{4L}, \frac{3v}{4L}, \frac{5v}{4L}, \frac{7v}{4L}$

Answer: B

23 fringe separation $= \frac{\lambda D}{d} = \frac{(600 \times 10^{-9})(2.5)}{0.001} = 3.75 \times 10^{-3} \text{ m}$